

Neutron Star Equations of State Through Relativistic Mean Field Theory

N. Mantilla-Molina ¹, L. M. Becerra ², J. F. Rodríguez-Ruiz ³ and L. A. Núñez ¹

¹Escuela de Física, Universidad Industrial de Santander, Bucaramanga, 680002, Santander, Colombia

²Centro Multidisciplinario de Física, Universidad Mayor, Santiago de Chile, 8580745, Chile

³Departamento de Física, Universidad Antonio Nariño, Bogotá D.C. 110231, Colombia

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Resumen

En este trabajo se estudió la ecuación de estado de la materia nuclear en condiciones de alta densidad, que describe la materia al interior de estrellas de neutrones, utilizando la Teoría Relativista de Campo Medio. El objetivo principal fue construir ecuaciones de estado para estrellas de neutrones que satisfagan las restricciones nucleares y astrofísicas actuales, y examinar la viabilidad de estrellas de neutrones masivas que puedan reproducir las observaciones más recientes. Se exploró el espacio de parámetros del modelo relativista de campo medio, analizando las correlaciones entre los acoplamientos de los campos mesónicos (σ , ω , ρ) junto con los parámetros de autointeracción escalar, y las propiedades macroscópicas estelares obtenidas mediante la integración de las ecuaciones de Tolman–Oppenheimer–Volkoff. Los resultados mostraron una jerarquía en la influencia de los parámetros: los acoplamientos isoescalares A_σ , A_ω y los términos de autointeracción b y c gobiernan la rigidez de la ecuación de estado y la masa máxima, mientras que el acoplamiento isovectorial A_ρ afecta el radio de la estrella y la energía de simetría. El modelo construido permitió encontrar configuraciones que satisfacen simultáneamente las restricciones nucleares y astrofísicas. Se obtuvieron estrellas con masas máximas de hasta $2.79 M_\odot$, compatibles simultáneamente con el componente secundario de GW190814 y radios para estrellas canónicas ($1.4 M_\odot$) de las observaciones NICER.

Corresponding author: nicolas2210707@correo.uis.edu.co

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1. Introduction

The matter in the interior of neutron stars (NSs) reaches densities several times the nuclear saturation density n_0 , under conditions that cannot be reproduced in terrestrial laboratories. The pressure–energy density relation in this regime, the equation of state (EOS) of dense matter, is the microphysical input that fixes macroscopic NS properties (mass, radius) through the Tolman–Oppenheimer–Volkoff (TOV) structure equations (Oppenheimer & Volkoff, 1939), providing a direct link between nuclear physics and astrophysical observation.

Relativistic mean-field theory (RMF) describes nuclear matter interacting through effective meson exchange within a covariant, relativistic framework (Walecka, 1974; Glendenning, 2000). Calibrating its couplings to empirical nuclear-matter properties alone does not yield a unique EOS. Different parameter sets can reproduce the same

nuclear data while predicting different NS masses and radii. Multi-messenger data, like precise pulsar-timing masses, NICER mass–radius measurements, and gravitational-wave events such as GW190814, provide independent constraints that partly addressed this degeneracy (Ascenzi et al., 2024). Here we systematically explore the RMF parameter space to identify the sets of parameters that simultaneously satisfy nuclear saturation properties and current mass/radius observations, and to determine the resulting maximum NS mass. The full derivation, additional results, and appendices are presented in Spanish in the original thesis (Mantilla-Molina et al., 2026).

2. Model and Methods

2.1. Equation of State

We describe the NS core as matter conformed by neutrons, protons and electrons, derived from the Lagrangian:

$$\begin{aligned}
\mathcal{L} = & \bar{\psi} \left[\gamma^\mu \left(i\partial_\mu - g_\omega \omega_\mu - \frac{1}{2} g_\rho \tau \cdot \rho_\mu \right) - (m - g_\sigma \sigma) \right] \psi \\
& + \frac{1}{2} (\partial_\mu \sigma \partial^\mu \sigma - m_\sigma^2 \sigma^2) - \frac{1}{3} b m (g_\sigma \sigma)^3 - \frac{1}{4} c (g_\sigma \sigma)^4 \\
& - \frac{1}{4} \omega_{\mu\nu} \omega^{\mu\nu} + \frac{1}{2} m_\omega^2 \omega_\mu \omega^\mu - \frac{1}{4} \rho_{\mu\nu} \cdot \rho^{\mu\nu} + \frac{1}{2} m_\rho^2 \rho_\mu \cdot \rho^\mu \\
& - g_\rho \rho_\mu \cdot [\rho_\nu \times \rho^{\nu\mu} + 2g_\rho (\rho^\mu \times \rho^\nu) \times \rho_\nu] \\
& + \bar{\psi}_e (i\gamma^\mu \partial_\mu - m_e) \psi_e,
\end{aligned} \tag{1}$$

in which nucleons interact via a scalar meson σ (attractive, with quartic self-interactions b, c), a vector meson ω (repulsive), and an isovector meson ρ (isospin symmetry), following the standard non-linear Walecka model (Walecka, 1974; Glendenning, 2000). In the mean-field approximation the meson fields are replaced by their ground-state expectation values, and the resulting algebraic equations are solved self-consistently under charge neutrality and β -equilibrium. The model has five free parameters: the coupling ratios $A_i \equiv (g_i/m_i)^2 m^2$ ($i = \sigma, \omega, \rho$; m the nucleon mass) and the self-interaction coefficients b and c . The core EOS is matched at sub-saturation densities to the standard BPS/BBP crust through a monotonic interpolation, ensuring causality ($c_s^2 < 1$) throughout the star.

2.2. Structure Equations and Parameter Exploration

For each EOS the stellar structure follows from the TOV equations,

$$\frac{dm}{dr} = 4\pi r^2 \rho, \tag{2}$$

$$\frac{dP}{dr} = -\frac{(\rho + P)(m + 4\pi r^3 P)}{r(r - 2m)}, \tag{3}$$

integrated outward from a central density $\rho(0) = \rho_c$ to the surface $P(R) = 0$, in geometrized units (Oppenheimer & Volkoff, 1939).

The five parameters are constrained through a systematic scan: for fixed b, c we search the A_σ - A_ω plane for the region reproducing the empirical saturation density $n_0 = 0.157 \pm 0.010 \text{ fm}^{-3}$ and binding energy per nucleon at saturation density $B/A = -15.97 \pm 0.40 \text{ MeV}$, verify that the compressibility modulus falls within its empirical range $K_0 = 230 \pm 40 \text{ MeV}$, and adjust A_ρ to satisfy the symmetry energy $a_{\text{sym}} = 31.6 \pm 2.7 \text{ MeV}$ and its slope $L_0 = 58.9 \pm 16 \text{ MeV}$ (Kumar et al., 2024; Drischler et al., 2024). Repeating this scan over a grid of (b, c) yields a family of nuclear-consistent EOS, for which we compute the maximum mass M_{max} , the canonical radius $R_{1.4}$, and the maximum compactness $C_{\text{max}} = M_{\text{max}}/R_{(M_{\text{max}})}$.

3. Results and Discussion

Figure 1 shows the sensitivity of each nuclear and stellar property to the five parameters. As expected from their physical roles, A_σ and A_ω dominate the saturation and stellar properties, while A_ρ is essentially decoupled from

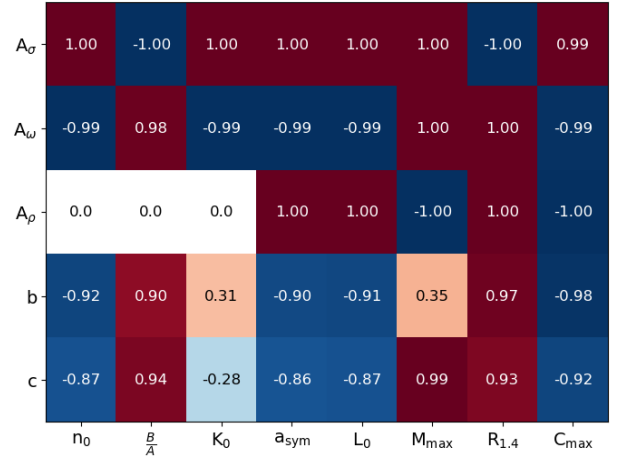


Figure 1. Pearson correlation coefficients between local variations of the five free parameters ($A_\sigma, A_\omega, A_\rho, b, c$) and the resulting nuclear and stellar properties, around the reference parameter set of Glendenning (2000). Values near +1 (−1) indicate a strong positive (negative) linear correlation.

symmetric-matter quantities and acts on a_{sym}, L_0 , and the stellar mass-radius. The self-interaction terms b and c have the weakest effect in regulating K_0 , and b has the smallest impact on M_{max} .

The nuclear-consistent parameter sets found span $M_{\text{max}} \simeq 2.0$ – $2.79 M_\odot$, $R_{1.4} \simeq 12.5$ – 13.7 km , and $C_{\text{max}} \simeq 0.28$ – 0.32 , confirming that nuclear saturation data alone do not select a unique EOS. Figure 2 compares the EOSs and mass-radius relations for the maximum-mass configurations obtained in several allowed regions. All models remain causal, while stiffer EOSs predict both larger maximum masses and larger radii. For reference, a free degenerate npe gas without strong interactions supports only $M_{\text{max}} \approx 0.7 M_\odot$, illustrating how essential the nuclear force is for reaching NS masses above $2 M_\odot$.

Nuclear interactions also reshape the microscopic composition of the star: the free npe gas keeps a neutron fraction n_n/n_B above $\sim 94\%$ up to $10^{16} \text{ g cm}^{-3}$, whereas every RMF configuration found here predicts a lower and more quickly varying n_n/n_B , reflecting the regulatory role of A_ρ on the proton-neutron asymmetry. Comparing the maximum-mass configuration of each valid region, the stiffer (higher M_{max}) sets are less neutron-rich at intermediate densities, but this ordering reverses above $n_B \simeq 0.4$ – 0.7 fm^{-3} , where the stiffest configurations become the most neutron-rich, linking the star's mass-supporting capacity to its isospin content.

The model with the highest mass found reaches $M_{\text{max}} = 2.79 M_\odot$ with $C_{\text{max}} = 0.316$, and exceeds the masses of the heaviest pulsars known, PSR J0740+6620 ($2.08 \pm 0.07 M_\odot$) (Fonseca et al., 2021) and PSR J0952-0607 ($2.35 \pm 0.17 M_\odot$) (Romani et al., 2022), as well as the secondary component of GW190814 ($2.59 M_\odot$) (LIGO et al., 2020). Its canonical radius, $R_{1.4} \simeq 12.8$ – 13.7 km , agrees with NICER

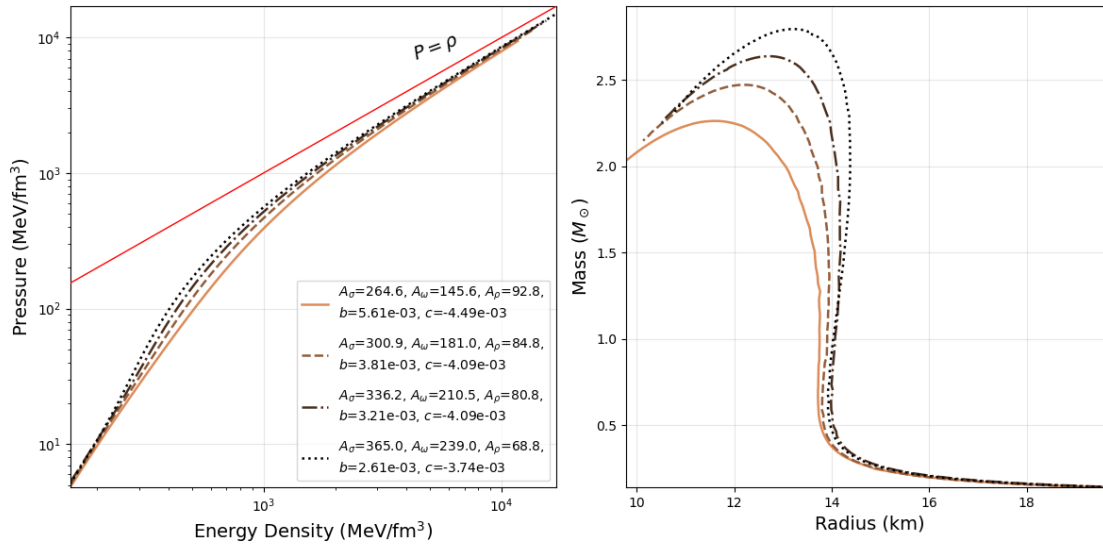


Figure 2. Equation of state (left) and mass-radius relation (right) for the maximum-mass configuration of each nuclear-consistent parameter region identified in this work. All configurations satisfy causality and the empirical nuclear constraints; stiffer equations of state yield larger maximum masses and canonical radii.

measurements of PSR J0030+0451 (Miller et al., 2019), PSR J0437-4715 (Choudhury et al., 2024), PSR J0740+6620 (Miller et al., 2021), and the canonical radius estimate for the GW190814 secondary (Biswas et al., 2021).

4. Conclusions

A relativistic mean-field EOS with non-linear scalar self-interactions can simultaneously satisfy empirical nuclear saturation constraints and current astrophysical mass and radius measurements, reaching maximum masses up to $2.79 M_{\odot}$ while remaining causal and physically allowed. The parameter exploration shows that A_{σ} and A_{ω} control the overall stiffness and maximum mass, whereas A_{ρ} regulates the symmetry energy and stellar radius nearly independently, providing practical guidance for future parameter fitting. The wide range of viable $(M_{\max}, R_{1.4})$ pairs remaining after imposing nuclear constraints confirms that laboratory data alone cannot fix the dense-matter EOS, highlighting the value of combining them with pulsar-timing, X-ray, and gravitational-wave observations. Natural extensions of this work include hyperonic and other exotic degrees of freedom, finite-temperature effects, and Bayesian parameter inference.

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